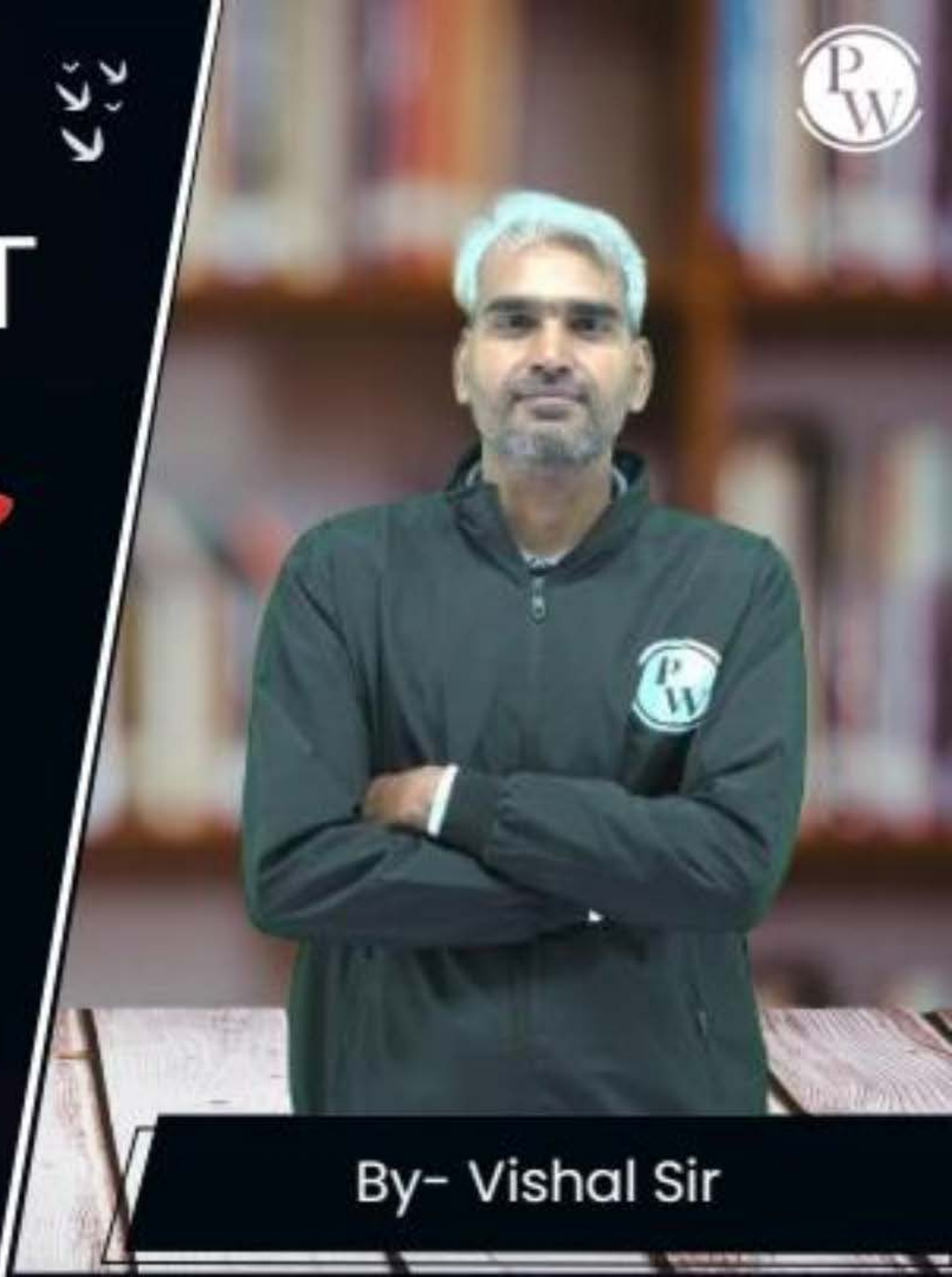
Computer Science & IT



Database Management System

Query Languages

Lecture No. 03



By- Vishal Sir

Recap of Previous Lecture



- **Topic** Join operation
- **Topic** Division operation

Slide

Topics to be Covered

Topic

Practice questions



Slide



Topic : Derived Relational Algebra operators

1. Intersection (∩)
2. Join Operations ("⋈")
3. Division Operation (÷)

* It is a set opⁿ, it can be performed only if relation are union compatible

⇓

Intersection is a derived relational algebra opⁿ, it is derived using "Set difference"

$R \cap S = R - (R - S)$



Topic : Join Operations

- Join operations are used to join relational tables based on some condition.

  * If we do not specify any condition, then it becomes Cross Join / Cross product.
  * Using cross join, all the tuples of the first relation will be joined with all the tuples of 2nd relation.



Topic : Types Join Operations

1) Inner join
   a. Theta join
   b. Equi join
   c. Natural join

{ Using inner join, we select only those tuples from both the relations that satisfy the join condition }

2) Outer join
   a. Left outer join
   b. Right outer join
   c. Full outer join

{ In outer join we can also select the tuples from one or both the relations, that failed the join Condⁿ }



Topic : Inner join

- Inner join, includes only those tuples that satisfy the join condition.



Topic : Theta join

The general case of JOIN operation is called a Theta join. It is denoted by symbol θ

i.e. $\bowtie_\theta$ — it may be any Cond^n.

- Theta join can use any conditions in the selection criteria.

$<$, $>$, $\leq$, $\geq$, $=$, $<>$ (Not equal)

✓ Student (S)

Sid	Sname
2	A
3	B
4	A

✓ Employee (E)

Eid	Ename
2	B
4	A
5	B

We can use any Condⁿ in theta join

Student ⋈ Employee

(Student.Sid < Employee.Eid)

it is the join Condⁿ

≡

$$\left(\sigma_{S.Sid < E.Eid}(S \times E)\right)$$

O/P :

No projection oph is used

∴ All attributes of "S×E" will be present in output.

S.Sid	S.Sname	E.Eid	E.Ename
2	A	4	A
2	A	5	B
3	B	4	A
3	B	5	B
4	A	5	B



Topic : Equijoin

{ Equi-join is a type of theta join where Cond^{n} is equality Condition }

An equijoin is a theta join using the equality operator.
In equijoin the condition is always equality condition.

- It is a type of theta join.

└ But it need not be on Common attributes

Student (S)

Sid	Sname
2	A
3	B
4	A

Employee (E)

Eid	Ename
2	B
4	A
5	B

Attribute need not be Common attribute

$$\text{Student} \bowtie_{(\text{Student.Sid} = \text{Employee.Eid})} \text{Employee}$$

Equality cond^n :. it can be Called Equi-join

$$\equiv \left(\sigma_{S.Sid = E.Eid} (S \times E) \right)$$

o/p :

S.Sid	S.Sname	E.Eid	E.Ename
2	A	2	B
4	A	4	A

Student (S)

Sid	Sname
2	A
3	B
4	A

Enroll (E)

Sid	Cid
2	C_1
4	C_2
5	C_3

Attribut may be Common attribute as well

Student $\bowtie$ (Student.Sid = Enroll.Sid) Enroll

Equi-join

$\equiv \left(\sigma_{S.Sid = E.Sid}(S \times E)\right)$

It is not natural join. In natural join we do not specify join condn.

O/P :

S.Sid	S.Sname	E.Sid	E.Cid
2	A	2	C_1
4	A	4	C_2

When we use Equi-join, Common attributes from both the relation will be Present in o/p

Note:- If we use theta join or Equi-join then we need to specify the join condition Explicitly. Which is not the case with natural join.

In natural join we do not specify any Condⁿ; because by default Condⁿ for natural join is Equality Condition on all Common attributes.



Topic : Natural join (⋈)

Natural join is performed based on the equality condition on all common attributes (column) between the relations. The name and type of common attributes must be same.
Additionally, a natural join removes the duplicate columns involved in the equality comparison so only one of each compared column remains

Student (Sid, Sname) & Enroll (Stu-id, Cid)
There is no common attribute between Student & Enroll w.r.t. Natural join

Natural Join ($\bowtie$)

Natural join is a derived relational algebra oph. which can be derived using three basic relational algebra oph. i.e; Cross product, Selection and Projection.

Derivation of Natural join

$R \bowtie S =$

Step-1 : obtain R x S

Step-2 : Select the tuples from R x S based on Equality Condn on all common attributes of R and S.

Step-3 : From the output of Step-2 project distinct attributes

Student (S)

Sid	Sname
2	A
3	B
4	A

Enroll (E)

Sid	Cid
2	C_1
4	C_2
5	C_3

$$\text{Student} \bowtie \text{Enroll} = \pi_{Sid, Sname, Cid}\left(\sigma_{Student.Sid = Enroll.Sid}\left(\text{Student} \times \text{Enroll}\right)\right)$$

No Condn is specified
↓
∴ Natural join

O/P =

"Sid" will be present only once in the o/p

Sid	Sname	Cid
2	A	C_1
4	A	C_2

eg: $R(A\ B\ C)$ & $S(B\ C\ D)$

$$R \bowtie S = \pi_{A,B,C,D}\left(\sigma_{\substack{R.B = S.B \\ \wedge \\ R.C = S.C}} (R \times S) \right)$$

Equality Condⁿ ✓ on all common attributes

eg. $R(\bar{A}, \bar{B}, \bar{C})$ & $S(\bar{D}, \bar{E})$ ✓

$R \bowtie S = \pi_{A,B,C,D,E}(\underbrace{\quad\quad}_{\text{No Common attributes b/w R \& S}}(R \times S)) \equiv R \times S$

∴ No selection Condⁿ.

↓

And hence all the tuples of R×S will get selected

And finally all attributes of R×S will be projected.

∴ O/p will be exactly same as R×S

If there is no common attribute between R and S, then Natural join of R and S will be equal to "R×S"

Student (S)

Sid	Sname
2	A
3	B
4	A

Employee (E)

Eid	Ename
2	B
4	A
5	B

$$\underset{(S)}{\text{Student}} \bowtie \underset{(E)}{\text{Employee}} \equiv S \times E$$

No Common attributes b/w Student & Enroll



Topic : Outer join

In an outer join, along with tuples that satisfy the join condition, we also include tuples that do not match the criteria either from left hand side relation, or from right hand side relation or both.



Topic : Left Outer Join (⟕)

$R \text{ ⟕ } S$ = All tuples of $R \bowtie S$, along with the tuples from left hand side relation {i.e. Rel^n R} that failed the join Cond^n are present.



Topic : Right Outer Join (⟖)

$R \rightarrow\!\!\!\bowtie S$: All tuples of $R \bowtie S$ along with the tuples of right hand side relation {i.e, Rel^n S} that failed the join condition are present



Topic : Full Outer Join (⟗)

R ⟗ S : All tuples of R ⋈ S along with the tuples from both side relation that failed the join condition are present

R =

	A	B	C
✓	1	2	3
✗	1	4	5

S =

	B	C	D
✓	2	3	4
✗	5	6	7
✗	4	7	9

R ⋈ S =

A	B	C	D
1	2	3	4

R ⟕ S =

A	B	C	D
1	2	3	4
1	4	5	NULL

R ⟖ S =

A	B	C	D
1	2	3	4
NULL	5	6	7
NULL	4	7	9

R ⟗ S =

A	B	C	D
1	2	3	4
1	4	5	NULL
NULL	5	6	7
NULL	4	7	9



Topic : Division (÷)

- Division operation is used whenever the query is with respect to every or all.

Eg. Consider following three relations.

Student (S)

Sid	Sname
S_1	A
S_2	A
S_3	B
S_4	C

Enroll (E)

Sid	Cid	fee
S_1	C_1	-
S_1	C_2	-
S_2	C_2	-
S_1	C_3	-
S_2	C_3	-
S_3	C_3	-

Course (C)

Cid	Cname
C_1	OS
C_2	COA
C_3	DBMS

Query:- Retrieve Sids of the students who have enrolled for all Courses.

Enroll (E)

Sid	Cid
S_1	C_1
S_1	C_2
S_2	C_2
S_1	C_3
S_2	C_3
S_3	C_3

$\Pi_{Sid}(E) \Rightarrow$ O/P =

Sid
S_1
S_2
S_3

This R.A. Expression Will produce the Sids of the Students Who have enroll for Some (at least) Course

Course (C)

Cid	Cname
C_1	OS
C_2	COA
C_3	DBMS

$\pi_{Cid}\ (C) \Longrightarrow$ O/P =

Cid
C_1
C_2
C_3

This R.A. Expression will produce Cids of all Courses.

* $\pi_{sid}(E)$ =

Sid
S_1
S_2
S_3

✓

* $\pi_{cid}(C)$ =

Cid
C_1
C_2
C_3

✓

$\underbrace{\pi_{sid}(E) \times \pi_{cid}(C)}$ ⇒ o/p =

This R.A. Expression will produce a universal relation, in which Every student who has Enrolled for some Courses is Combined with all Courses.

E.Sid	C.Cid
S_1	C_1
S_1	C_2
S_1	C_3
S_2	C_1
S_2	C_2
S_2	C_3
S_3	C_1
S_3	C_2
S_3	C_3

Student (S)

Sid	Sname
S_1	A
S_2	A
S_3	B
S_4	C

Enroll (E)

Sid	Cid	fee
S_1	C_1	
S_1	C_2	
S_2	C_2	
S_2	C_3	
S_3	C_3	
S_3	C_3	

Course (C)

Cid	Cname
C_1	OS
C_2	COA
C_3	DBMS

* Retrieve Sids of the students who have Enrolled for all Courses.

↳ We are actually looking for the Sids in the Enroll table which are associated with all the Cids of the Course table.

↳ desired result can be produced by following relational algebra Expression

$$\pi_{Sid,Cid}(Enroll) \div \pi_{Cid}(Course)$$

it will produce Sids from Enroll table Which are associated with all Cids of Course table

* Derivation of division(÷) operation using basic relational algebra operation

Student (S)

Sid	Sname
S_1	A
S_2	A
S_3	B
S_4	C

Enroll (E)

Sid	Cid	fee
S_1	C_1	
S_1	C_2	
S_2	C_1	
S_1	C_3	
S_2	C_2	
S_3	C_3	

Course (C)

Cid	Cname
C_1	OS
C_2	COA
C_3	DBMS

Student Student

$$\pi_{sid,cid}(E) \div \pi_{cid}(C)$$

$$\equiv \left[\pi_{sid}(E) - \pi_{sid}\left[(\pi_{sid}(E) \times \pi_{cid}(C)) - \pi_{sid,cid}(E)\right]\right]$$

Sid
S_1
S_2
S_3

Sid
S_2
S_3

Sids of students who did not enroll for all courses

Sid
S_1
S_2
S_3

Cid
C_1
C_2
C_3

Sid	Cid
S_1	C_1
S_1	C_2
S_1	C_3
S_2	C_1
S_2	C_2
S_2	C_3
S_3	C_1
S_3	C_2
S_3	C_3

Sid	Cid
S_2	C_1
S_3	C_1
S_3	C_2

We will get information of the students who did not enroll for all courses, along with the Cids of courses for which they did not enroll

Sid
S_1

Sids of students who enrolled for all courses

Sid	Cid
S_1	C_1
S_1	C_2
S_1	C_3
S_2	C_1
S_2	C_2
S_2	C_3
S_3	C_1
S_3	C_2
S_3	C_3

- $\pi_{A,B,C}(R) \div \pi_{C}(S)$ => o/p will be (A,B) values from relation 'R' which associate with all 'C' values of relations S

- $\pi_{A,B,C}(R) \div \pi_{B,C}(S)$ => o/p will 'A' values from relation R that are associated with all (BC) values of relation S.

$$\pi_{sid,cid}(E) \div \pi_{cid}(C)$$

$$\equiv \pi_{sid}(E) - \pi_{sid}\left[\pi_{sid}(E) \times \pi_{cid}(C) - \pi_{sid,cid}(E)\right]$$

① ⑧ ⑦ ② ④ ③ ⑥ ⑤



Consider the following relational tables:

- Supplier (Sid, Sname, Rating)
- Parts (Pid, Pname, Color)
- Catalog (Sid, Pid, Cost)

Suppliers

Sid	Sname	Rating
S_1	A	3
S_2	A	5
S_3	B	7
S_4	C	0

Parts

Pid	Pname	Color
P_1	XYZ	Red
P_2	PQR	Green
P_3	XYZ	Red

Catalog

Sid	Pid	Cost
S_1	P_1	10
S_2	P_2	20
S_3	P_3	30

* Retrieve Sids of suppliers who supplied at least one part.

π_{sid} (Catalog)

⇒ o/p

Sid
s_1
s_2
s_3

* Retrieve Sids of all the suppliers

π_{sid} (Supplier)

⇒ o/p

Sid
s_1
s_2
s_3
s_4

Q:- Retriere Sids of the suppliers whose rating is more than 5

$$\Pi_{Sid}\left(\sigma_{Rating>5}(Supplier)\right)$$

o/p =

Sid
S_3

o/p =

Sid	Sname	Rating
S_3	B	7

Q. Retrieve Sids of Suppliers who supplied some red color parts.

Catalog

Sid	Pid	Cost
S_1	P_1	10
S_2	P_2	20
S_3	P_3	30

Parts

Pid	Pname	Color
P_1	XYZ	Red
P_2	PQR	Green
P_3	XYZ	Red

① $\pi_{C.Sid}\left(\sigma_{(C.Pid = P.Pid \wedge P.Color = 'Red')}\left(\rho_C(Catalog) \times \rho_P(Parts)\right)\right)$

⇓

o/p =

Sid
S_1
S_3

C.Sid	C.Pid	C.Cost	P.Pid	P.name	P.Color
S_1	P_1	10	P_1	XYZ	Red
S_1	P_1	10	P_2	PQR	Green
S_1	P_1	10	P_3	XYZ	Red
S_2	P_2	20	P_1	XYZ	Red
S_2	P_2	20	P_2	PQR	Green
S_2	P_2	20	P_3	XYZ	Red
S_3	P_3	30	P_1	XYZ	Red
S_3	P_3	30	P_2	PQR	Green
S_3	P_3	30	P_3	XYZ	Red

Catalog

Sid	Pid	Cost
S_1	P_1	10
S_2	P_2	20
S_3	P_3	30

Parts

Pid	Pname	Color
P_1	xyz	Red
P_2	PQR	Green
P_3	xyz	Red

(2) $\pi_{Sid}\left(\sigma_{Color='Red'}(C \bowtie P)\right)$

C.Sid	C.Pid	C.Cost	P.Pid	P.name	P.Color
S_1	P_1	10	P_1	xyz	Red
S_1	P_1	10	P_2	PQR	Green
S_1	P_1	10	P_3	xyz	Red
S_2	P_2	20	P_1	xyz	Red
S_2	P_2	20	P_2	PQR	Green
S_2	P_2	20	P_3	xyz	Red
S_3	P_3	30	P_1	xyz	Red
S_3	P_3	30	P_2	PQR	Green
S_3	P_3	30	P_3	xyz	Red

Catalog

Sid	Pid	Cost
S_1	P_1	10
S_2	P_2	20
S_3	P_3	30

Parts

Pid	Pname	Color
P_1	XYZ	Red
P_2	PQR	Green
P_3	XYZ	Red

(3) $\uparrow_{Catalog.Sid}\left(\sigma_{Catalog.Pid = Parts.Pid}\left(Catalog \times \left(\sigma_{Color='Red'}(Parts)\right)\right)\right)$

(4) $\pi_{Sid}\left(Catalog \bowtie \left(\sigma_{Color='Red'}(Parts)\right)\right)$

Q. Retrieve Sids of Suppliers who supplied some red color parts.

1000 1000

① $\pi_{C.sid}\left(\sigma_{C.Pid=P.Pid \wedge P.Color='Red'}(C \times P)\right)$ — $C \times P$: 1000000

② $\pi_{sid}\left(\sigma_{Color='Red'}(C \bowtie P)\right)$

Less efficient

50,000

③ $\pi_{C.sid}\left\{\sigma_{C.Pid=P.Pid}\left(C \times \left(\sigma_{Color='Red'}(P)\right)\right)\right\}$ — C: 1000; $\sigma_{Color='Red'}(P)$: 50

④ $\pi_{sid}\left\{C \bowtie \left(\sigma_{Color='Red'}(P)\right)\right\}$

More efficient

#Q. Retrieve Sid of the suppliers who have supplied some Red or some Green color parts.

Catalog

Sid	Pid	Cost
S_1	P_1	10
S_2	P_2	20
S_3	P_3	30

Parts

Pid	Pname	Color
P_1	XYZ	Red
P_2	PQR	Green
P_3	XYZ	Red

C.Sid	C.Pid	C.Cost	P.Pid	P.name	P.Color
S_1	P_1	10	P_1	XYZ	Red
S_1	P_1	10	P_2	PQR	Green
S_1	P_1	10	P_3	XYZ	Red
S_2	P_2	20	P_1	XYZ	Red
S_2	P_2	20	P_2	PQR	Green
S_2	P_2	20	P_3	XYZ	Red
S_3	P_3	30	P_1	XYZ	Red
S_3	P_3	30	P_2	PQR	Green
S_3	P_3	30	P_3	XYZ	Red

Q. Retrieve Sids of Supplies who supplied some red or some green Color Parts.

① $\pi_{C.sid}\left(\sigma_{(C.Pid=P.Pid \wedge (P.Color='Red' \vee P.Color='Green'))}(C \times P)\right)$

② $\pi_{sid}\left(\sigma_{(Color='Red' \vee Color='Green')}(C \bowtie P)\right)$

③ $\pi_{C.sid}\left\{\sigma_{C.Pid=P.Pid}\left(C \times \left(\sigma_{(Color='Red' \vee Color='Green')}(P)\right)\right)\right\}$

④ $\pi_{sid}\left\{C \bowtie \left(\sigma_{(Color='Red' \vee Color='Green')}(P)\right)\right\}$

#Q. Retrieve Sid of the suppliers who have supplied some Red and some Green color parts.

Q. Retrieve Sids of Suppliers who supplied some red and some green Color Parts.

① $\pi_{C.sid}\left(\sigma_{(C.Pid = P.Pid \wedge (P.Color = 'Red' \wedge P.Color = 'Green'))}(C \times P)\right)$

② $\pi_{sid}\left(\sigma_{(Color = 'Red' \wedge Color = 'Green')}(C \bowtie P)\right)$

③ $\pi_{C.sid}\left\{\sigma_{C.Pid = P.Pid}\left(C \times \left(\sigma_{(Color = 'Red' \wedge Color = 'Green')}(P)\right)\right)\right\}$

④ $\pi_{sid}\left\{C \bowtie \left(\sigma_{(Color = 'Red' \wedge Color = 'Green')}(P)\right)\right\}$

All are wrong.

⇓

All four queries will always produce Empty relation

⇓

In a single tuple, Color will either be 'Red' or 'Green' but Can never be both

Q. Retrieve Sids of Suppliers who supplied some red and some green color parts

① $\pi_{C.sid}\left(\sigma_{C.Pid=P.Pid \wedge P.Color='Red'}(C \times P)\right) \cap \pi_{C.sid}\left(\sigma_{C.Pid=P.Pid \wedge P.Color='Green'}(C \times P)\right)$

② $\pi_{sid}\left(\sigma_{Color='Red'}(C \bowtie P)\right) \cap \pi_{sid}\left(\sigma_{Color='Green'}(C \bowtie P)\right)$

Q. Retrieve Sids of Suppliers who supplied some red and some green color parts

③ $\pi_{C.Sid}\{\sigma_{C.Pid=P.Pid}(C \times (\sigma_{Color='Red'}(P)))\} \cap \pi_{C.Sid}\{\sigma_{C.Pid=P.Pid}(C \times (\sigma_{Color='Green'}(P)))\}$

④ $\pi_{Sid}\{C \bowtie (\sigma_{Color='Red'}(P))\} \cap \pi_{Sid}\{C \bowtie (\sigma_{Color='Green'}(P))\}$

Q. Retrieve Sids of Supplies who supplied some red or some green Color parts

① $\pi_{C.sid}\left(\sigma_{C.Pid=P.Pid \wedge P.Color='Red'}(C \times P)\right) \cup \pi_{C.sid}\left(\sigma_{C.Pid=P.Pid \wedge P.Color='Green'}(C \times P)\right)$

② $\pi_{sid}\left(\sigma_{Color='Red'}(C \bowtie P)\right) \cup \pi_{sid}\left(\sigma_{Color='Green'}(C \bowtie P)\right)$

Q. Retrieve Sids of Suppliers who supplied some red or some green Color Parts

③ $\pi_{C.Sid}\left\{\sigma_{C.Pid = P.Pid}\left(C \times \left(\sigma_{Color='Red'}(P)\right)\right)\right\} \cup \pi_{C.Sid}\left\{\sigma_{C.Pid = P.Pid}\left(C \times \left(\sigma_{Color='Green'}(P)\right)\right)\right\}$

④ $\pi_{Sid}\left\{C \bowtie \left(\sigma_{Color='Red'}(P)\right)\right\} \cup \pi_{Sid}\left\{C \bowtie \left(\sigma_{Color='Green'}(P)\right)\right\}$

- Another approach to retrieve Sids of Suppliers who supplied some red and some green Color parts.

Catalog

Sid	Pid	Cost
S_1	P_1	10
S_2	P_2	20
S_3	P_3	30
S_1	P_2	30

Parts

Pid	Pname	Color
P_1	xyz	Red
P_2	PQR	Green
P_3	xyz	Red

$(C_1 \times P_1) \times (C_2 \times P_2)$

C_1.Sid	C_1.Pid	C_1.Cost	P_1.Pid	P_1.Pname	P_1.Color
S_1	P_1	10	P_1	xyz	Red
S_1	P_1	10	P_2	PQR	Green
S_1	P_1	10	P_3	xyz	Red
S_2	P_2	20	P_1	xyz	Red
S_2	P_2	20	P_2	PQR	Green
S_2	P_2	20	P_3	xyz	Red
S_3	P_3	30	P_1	xyz	Red
S_3	P_3	30	P_2	PQR	Green
S_3	P_3	30	P_3	xyz	Red

C_2.Sid	C_2.Pid	C_2.Cost	P_2.Pid	P_2.Pname	P_2.Color
S_1	P_1	10	P_1	xyz	Red
S_1	P_2	30	P_2	PQR	Green
S_1	P_1	10	P_3	xyz	Red
S_2	P_2	20	P_1	xyz	Red
S_2	P_2	20	P_2	PQR	Green
S_2	P_2	20	P_3	xyz	Red
S_3	P_3	30	P_1	xyz	Red
S_3	P_3	30	P_2	PQR	Green
S_3	P_3	30	P_3	xyz	Red

- Another approach to retrieve Sids of Suppliers who supplied some red and some green Color parts.

$$\pi_{C_1.Sid}\Big\{\sigma\big(C_1.Sid = C_2.Sid \wedge (C_1.Pid = P_1.Pid \wedge P_1.Color = 'Red') \wedge (C_2.Pid = P_2.Pid \wedge P_2.Color = 'Green')\big)\{(C_1 \times P_1) \times (C_2 \times P_2)\}\Big\}$$

We are using $C_1 \times P_1$ w.r.t 'Red' Color parts

We are using $C_2 \times P_2$ w.r.t. Green Color parts.

#Q. Retrieve Sid of the suppliers who have supplied all parts.

#Q. Retrieve Sid of the suppliers who have supplied all Red color parts.

#Q. Retrieve Sid of the suppliers who have supplied at least two parts.

#Q. Retrieve Sid of the suppliers who have supplied exactly one part.

#Q. Retrieve Sid of the suppliers who have supplied at most one part.

#Q. Retrieve Sid of the suppliers who have supplied at least three parts.

#Q. Retrieve Sid of the suppliers who have supplied most expensive parts.



2 mins Summary

Topic Practice questions

Slide

THANK - YOU